MIL-R-45677(Ord)
25 July 1960
SUPERSEDING
FA (FC) - PD - 118
14 January 1959

MILITARY SPECIFICATION

RANGEFINDER, FIRE CONTROL, M17C

1. SCOPE

1.1 This specification covers one type of coincidence image instrument known as Rangefinder M17C for determining range measurement in meters in the 90MM Gun Tank M48A2C and 105MM Gun Main Battle Tank M60 Fire Control Systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components of Fire Control Instruments: General Specification Governing the Manufacture, Assemtly and Inspection of
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property

DRAWINGS

Ordnance Corps

F8620800	Rangefinder, Fire Control, M17C
LSQAP 8620800	List of Supplementary Quality Control Assurance Provisions
IEL 8620800	Index to List of Inspection Equipment

FSC 1240

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(Copies of specifications, standards, drawings, and publications required by the contractor in connection with this specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production Model. - At the beginning of production one rangefinder, representative of production shall be made available to the government to determine if the rangefinder meets all requirements of this specification.

3.2 Materials. - Materials shall be in accordance with the drawings, general specifications and material specifications forming part of this specification.

3.3 Construction. - The rangefinder shall be manufactured in accordance with Ordnance Corps Drawing F8620800 Rangefinder, Fire Control, M17C and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the government.

NOTE: No rangefinder shall be accepted for testing which has not been previously inspected, prior to assembly of the main hanger bearings, for precise positioning of the Adapter Link Assembly 8201530 as required by Drawing F8620801-4. Two production end housings shall be identified as components of each rangefinder by stenciling the rangefinder serial number, and end housing position (left and right) on the exterior of each end housing. Under no circumstances will temporary substitution of other end housings be permitted for completion of requirements herein or for shipment of the rangefinder.

3.4 General specifications. - All requirements as specified in section 3 of MIL-F-13926 and MIL-O-13830 shall apply, except those requirements not logically related to the rangefinder.

3.5 Pre-vibration. -

3.5.1 Orientation. - The rangefinder prior to vibration shall be set-up in a horizontal position as shown on the Assembly View Drawing F8620800 and all instructions thereon, and the following conditions shall be established prior to testing:

NOTE: All coincident readings of the target shall be made in the center five mils of the true field unless otherwise required.

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(1) The line through the centers of the two mounting bearings horizontal.

(2) Identified production end housings in place.

(3) Level bubble centered within width of a vial graduation.

(4) Calibrating reticle illuminated.

(5) Occuluder set at mid-index.

(6) Filters out of the optical system.

(7) A target provided at 1500 meters plus or minus 100 meters and the range scale set to agree with the range chosen.

(8) Set the vertical calibration knob at mid-point of total travel within 10 degrees and slip the scale to aline indices. Establish vertical coincidence of the target image.

(9) Set the horizontal calibration knob at mid-point of total travel within one division on the scale, slip the scale to zero index, make a series of five horizontal coincidence readings on the target image. Rotate the knob to the median reading.

(10) Set the deflection (azimuth) and elevation boresight knobs and the auxiliary deflection (azimuth) and elevation boresight knobs at mid-point of total travel within 0.1 mil graduation. Slip the deflection (azimuth) scales to the three (3) position and slip the elevation scales to the two (2) position.

(11) The left boresight reticle shall be set to bring the line of sight 1 mil below the horizontal, and 1 mil to the right of the perpendicular to a center line through the center of the mounting bearing within .05 mil (reticle line width) and the knobs shall then be locked.

(12) The right boresight reticle shall set so that it's line of sight coincides with the aiming point of the left reticle within .05 mil (reticle line width) and the knobs shall then be locked.

3.5.2 Knob travel (calibration).

3.5.2.1 The horizontal calibration knob shall read within 7 units of error (UOE) of its mid-point of total travel. If requirement is met, the scale shall then be slipped to the zero index.

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3.5.2.2 The vertical knob shall be within 30 degrees of its previously established mid-point. If requirement is met slip the scale to aline indices.

3.5.2.3 The scale readings on all boresight knobs shall be within 0.7 mil (knob graduation) of their mid-points. If requirement is met slip the scales to three (3) position and two (2) position as indicated in 3.5.1 (10).

3.5.3 Calibration reticle setting accuracy.

3.5.3.1 The calibrating reticle (lower) that is fixed with respect to the external field shall lie in its entirety between 5 and 12 mils above the center of the boresight cross and the vertical line of the calibrating reticle shall be within 3 mils left and 1 mil right of the vertical center line of the sighting reticle. The halving knob shall then be adjusted to position the movable (upper bar) of the calibrating reticle, such that with internal correction system (ICS) knob movement, the upper bar will pass within one reticle line thickness of the lower reticle, yet maintaining a recognizable separation between the two reticle bars during movement.

3.5.3.2 Five reticle coincidence setting's shall be made with the ICS adjustment. The median of the base five readings shall read within 7 UOE of the (25) position.

3.6 Vibration. - After meeting all the above requirements and with the end housing assemblies removed, (removal of end housings optional) the rangefinder shall be vibrated at an amplitude of not less than 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second for a duration of 5 minutes, plus or minus 15 seconds.

3.6.1 Production end housing assemblies. - The end housing assemblies shall be vibrated as separate units at an amplitude of not less than 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second for a duration of 5 minutes, plus or minus 15 seconds.

3.7 Post-vibration.

3.7.1 Post-vibration optical re-check. - With the identified end housing assemblies replaced on the rangefinder, and the rangefinder restored to the conditions of 3.5.1 (1) thru (7), the rangefinder shall meet the following requirements:

3.7.1.1 Alinement of sighting reticles. - The line of sight of the left telescope through the boresight mark shall not have changed by more than 0.25 mil in deflection (azimuth) and 0.5 mil in elevation from the value introduced in 3.5.2.3. The line of sight of the right telescope shall coincide with the aiming point of the left telescope within a maximum angular separation of 0.1 mil.

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3.7.1.2 Image. - The target image shall not have changed coincidence by more than 0.5 mil in vertical calibration or more than 25 UOE in horizontal calibration as measured against a calibrated external target using the median of five coincident settings made with the horizontal calibration knob.

3.7.1.3 Coincidence reticle setting. - The space between the two calibrating reticles shall not decrease past a distinguishable spacing and shall not increase in excess of 0.2 mil.

3.7.1.4 Set the horizontal calibration knob at the new median setting, then using the ICS knob, obtain a median from five ICS settings of the calibrating reticle. This median shall not differ by more than 5 UOE from the median reading established in 3.5.3.2 prior to vibration.

3.7.1.5 Subsequent to vibration and meeting the requirements of 3.7.1.1 to 3.7.1.4 inclusive, the rangefinder shall meet all requirements of 3.7.2 to 3.7.17 inclusive.

NOTE: Should the rangefinder fail to meet all requirements of 3.7.1, the contractor may request that the identified end housing assemblies be removed and replaced, using only the same identified end housings, in order to insure that the installation of the end housings was not at fault. Should the results of the second installation be at variance with the first, one or more additional installations may be made until two consecutive installations repeat within required limits, after which the instrument shall be considered acceptable to this stage (3.7.1).

3.7.1.6 End housing assemblies. - Subsequent to vibration, end housing assemblies shall comply with the requirements of Ordnance Drawing F8585506.

3.7.2 Inspection for dirt and bright spots.

3.7.2.1 Inspection for dirt. - The maximum real diameter of any 1 defect shall not exceed 0.2 mil in the outer 22 1/2 degrees, and 0.05 mil in the central 17 1/2 degrees of the apparent field as viewed at the eyepiece. The sum of the maximum real diameter of dirt and pinholes in the rangefinder (less end housings) shall not exceed 1 mil. Each end housing assembly shall meet the cleanliness requirements of MIL-O-13830. Dirt on the optical surfaces of the end housing shall not exceed in size and number the limitations for size and number of digs permitted for each surface as specified on applicable Ordnance drawings.

3.7.2.2 Bright spots. - Bright spots (pinholes) in reticle shall not exceed in size and number the limitations for size and number of digs permitted for that surface as specified on the applicable Ordnance drawings.

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3.7.3 Parallax

3.7.3.1 Parallax of sighting reticles. - Utilizing a target at 1500 meters plus or minus 100 meters and with the range scale set to agree with the range chosen, parallax in the right and left systems at the optical axis (boresight cross) shall not exceed 0.1 mil, nor shall there be more than 0.1 mil parallax between reticles.

3.7.3.2 Parallax of the calibrating reticles. - The median of five ICS readings taken with the eye centered at 0.030 inch to the left, and the median of five readings taken with the eye centered at 0.030 inch to the right of the center of the common exit pupil shall not differ by more than 6 UOE as measured on the ICS knob.

3.7.3.3 Parallax of the target images. - Utilizing a target at 1500 meters plus or minus 100 meters the median of five coincidence readings taken with the eye centered 0.030 inch to the left of the center of the exit pupil shall not differ from the median of five coincidence readings taken with the eye centered 0.030 inch to the right of the center of the exit pupil by more than 6 UOE as read on the horizontal calibration knob.

3.7.4 Exit pupil. - The minimum diameter of the common exit pupil when viewed in the direction of the optical axis shall be 0.110 inch (2.79 millimeters).

3.7.5 Eyepiece focus. -

3.7.5.1 The eyepiece focus shall be adjustable over the range of plus or minus 3 diopters as read on the diopter scale.

3.7.5.2 The zero of the diopter scale shall indicate zero diopters within 0.25 diopter as measured against the left target image.

3.7.6 Definition. - The resolving power on the optical axis of each optical system shall be 6 seconds of arc or better.

3.7.7 Light transmission. - The difference in light transmission between the two optical systems shall not exceed 30 percent of the higher transmission value.

3.7.8 Sighting reticles. -

3.7.8.1 Accuracy of sighting reticles. - The angular value of the total horizontal lines and spaces of the reticles, in eighter field of view shall be 40 mils plus or minus 1.0 mil.

3.7.8.2 Plumb travel. - The left boresight mark when projected into the eyepiece shall not deviate from a plumb line by more than 0.2 mil when the rangefinder is elevated from 8 degrees depression to 19 degrees elevation, starting with the line of sight in coincidence with the plumb line.

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3.7.8.2.1 The rangefinder shall display unrestricted movement through an excursion of at least 22 degrees in elevation (above a horizontal line of sight) and 15 degrees in depression (below a horizontal line of sight).

3.7.8.3 Parallelism of sighting reticle and image. - When either sighting reticle is projected into the eyepiece the longest horizontal reticle line (40 mil line) shall be parallel to the image of a horizontal target line within 43 minutes of arc (43 minutes of arc is equivalent to an angle subtended by a 0.5 mil block at one end of the reticle line when the other end is in coincidence with the target line).

3.7.8.4 Tilt variation between reticles. - When the right reticle is superimposed on the left reticle the variation, between the reticles, as measured at the 40 mil reticle line, shall not exceed the width of a reticle line at one extremity of the 40 mil reticle lines when the other extremity of both 40 mil reticle lines are coincident.

3.7.9 Coincidence reticles.

3.7.9.1 Parallelism of calibrating reticles. - Each calibrating reticle shall be parallel to the image of a plumb line within 1 degree of arc.

3.7.9.2 Reversing reticles. - Utilizing the halving knob, the upper and lower reticle bars shall be reversed by putting the upper bar below. The median of five reticle coincidence readings on the ICS knob shall not differ by more than 5 UOE from the median established in 3.7.1.4.

3.7.9.3 Vertical movement accuracy. - The rangefinder shall be reset to the requirements of 3.7.1.4. When the vertical calibration knob is rotated to its maximum counterclockwise (CCW) position and the reticle returned to its original position by means of the halving knob the median of 5 deflection (azimuth) readings (UOE) of the calibrating reticle using the ICS knob shall not differ by more than 10 UOE from the median of 5 deflection (azimuth) readings obtained when the vertical calibration knob is rotated to maximum clockwise (CW) position and the reticle returned to its original position by means of the halving knob.

3.7.9.4 Horizontal calibration. - From the setting of 3.7.1.4 the horizontal calibration knob shall be rotated from the median setting (coincidence position of left and right images) to plus 40 and minus 40. The range for each extreme shall be equivalent to the corresponding change in the number of UOE on the knob scale within 10 percent.

3.7.9.5 ICS calibration. - From the settings of 3.7.1.4 offset the ICS knob 20 UOE CCW, and re-establish reticle coincidence by rotation of the horizontal calibration knob. Then rotate the ICS knob 40 UOE CW and re-establish reticle coincidence by rotation of the horizontal calibration knob. The excursion of the horizontal calibration knob shall be 40 UOE plus or minus 10 UOE as measured on the knob scale.

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3.7.10 Filters. - The introduction of filters into the optical system shall not cause more than 1 UOE change in horizontal calibration nor more than 0.1 mil in vertical calibration in coincidence.

3.7.11 Compensator elevation error. - When the rangefinder is actuated to cover the range span from 500 to 4400 meters the elevation of the two target images shall not change with respect to each other by more than 0.1 mil.

3.7.12 Range accuracy. - With the rangefinder set as in 3.7.1.4, the indicating device for rotational position of the output shaft shall be set to the values computed from the formula: P=23.454 Log 10R - 58.577, where "P" is the shaft position in turns and "R" is the target range in meters. At 1500 meters on the range scale, P=15.92.

3.7.12.1 The range of rotation of the output shaft shall be from P= 3.76 plus or minus 0.05 to P=27.25 plus or minus 0.20 turns, where the turn counter is adjusted in accordance with 3.7.12.

3.7.12.2 When ranging on targets at the ranges specified in table I the number of turns of the output shaft for each range shall not differ from the "P" value computed from the formula given in 3.7.12 in excess of the unit of error tolerances given in table I. For the computation of UOE with respect to the amount of turns of the output shaft use the relationship that 0.000029R (R is range in meters) corresponds to 1 UOE change in ranging. (This formula should not be used for other applications in this specification for values in excess of 5 UOE.

Table I Tolerance on Position of Rangefinder Output Shaft

Exact range in meters within spread	Range tolerance in UOE (Plus or Minus)
500 to 540	11
650 to 750	5
900 to 1100	2
1400 to 1600	0
2200 to 2800	2
3800 to 4400	2

3.7.12.3 The readings of the range scale shall agree with the positions taken by the output shaft when checked to the requirements of table II.

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Table II Tolerance on Reading of Range Scale

Position of output shaft	Reading of range scale	Tolerance on range scale reading (Plus or Minus)
520 meters	520 meters	5 meters
1000 meters	1000 meters	5 meters
1500 meters	1500 meters	0 meters
2500 meters	2500 meters	10 meters
4100 meters	4100 meters	20 meters

3.7.13 Field errors. - From the median setting established in 3.7.1.4, with the target in the center of the field (boresight cross) at 1500 (plus or minus 100 meters), the target shall be brought into coincidence at a point 20 mils (plus or minus 2 mils) above, below, right and left, of the center of the field, (boresight cross). The median of 5 coincidence readings at each point shall not differ from the median UOE reading obtained at the center of the field by more than 6 UOE.

3.7.14 Base length. - The base length of the rangefinder measured optically, when identified production end housing assemblies are installed, shall be 79 inches plus or minus 0.15 inch (2.0059 meters plus or minus 3.81 millimeters).

3.7.15 Knob travel. - All knobs shall have sufficient movement to cover the entire range of their respective scales.

3.7.16 Boresight knobs. -

3.7.16.1 Elevation. - Rotation of the elevation boresight knobs shall cause the line of sight to move at least 5 1/2 mils in depression and 5 1/2 mils in elevation from the nominal position established in 3.5.1 (11) and (12).

3.7.16.2 Deflection (azimuth). - Rotation of the deflection (azimuth) boresight knobs shall cause the line of sight to move at least 7 mils deflection each side of the nominal position established in 3.5.1 (11) and (12).

3.7.16.3 Backlash. - Backlash in the boresight knobs shall not exceed 0.05 mil in the line of sight as read on the knob scales.

3.7.16.4 Locks. - Actuation of the locks of the deflection (azimuth) and elevation boresight knobs shall not cause more than 0.05 mil shift

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in the line of sight and not more than 0.05 mil rotation of the boresight knob, as measured on the knob scale.

3.7.16.5 Lock retention. - When the locks of the boresight knobs are in the locked positions, the knob shall not move when a torque of 12 pound inches is applied to the knob.

3.6.17 Calibration knobs.

3.7.17.1 Backlash. - Backlash in the ICS knob shall not exceed 1/2 UOE and backlash in the horizontal calibration knob shall not exceed 1 UOE as read on the knob scales. Backlash in the halving knob as measured on the calibrating reticle shall not exceed 0.5 mils and backlash in the vertical calibration knob as read on the target reticle shall not exceed 0.25 mils.

3.7.17.2 Locks. - Locking of the horizontal and vertical calibration knobs by setting the knobs in the serration detent shall not introduce deflection (azimuth) error in excess of 1/4 UOE or elevation error in excess of the width of the index line, as read on the knob scales.

3.8 Operability. - When the rangefinder is exposed to the ambient temperatures of plus 60 degrees to plus 90 degrees fahrenheit (°F), of table III, the maximum torque required for operation of the controls shall not exceed the values specified.

Table III Operation at Room and Environmental Temperature

Control	Operability +60°F to + 90°F	Environmental -25°F	Environmental +150°F
Range Knob	70 oz. in.	16 lb. in.	70 oz. in.
Deflector (azimuth) and elevation boresight knobs	60 oz. in.	10 lb. in.	60 oz. in.
Horizontal & vertical calibration knobs (pull out force)	48 oz.	10 lb.	48 oz.
Horizontal & vertical calibration knob (rotation)	60 oz. in.	10 lb. in.	60 oz. in.
Halving knob	60 oz. in.	10 lb. in.	60 oz. in.
ICS knob	60 oz. in.	10 lb. in.	60 oz. in.
Diopter adjustment	30 oz. in.	10 lb. in.	30 oz. in.
Rheostat knob	30 oz. in.	8 lb. in.	30 oz. in.
Light switches	*m. o.	m.o.	m. o.
Computer switch	m. o.	m.o.	m. o.
Boresight & calibration locks	m. o.	m.o.	m. o.
Filter lever	m. o.	m.o.	m. o.
Occluder knob	m. o.	m.o.	m. o.

*m. o. - mechanically operable

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3.9 Sealing. - The rangefinder body and end housings shall show no evidence of leakage in excess of the following when subjected to an internal pressure of 6 inches of mercury at room temperature (60 degrees to 90 degrees) for a period of twelve hours.

Rangefinder body	0.3 inch mercury
End housing	0.1 inch mercury

3.9.1 Flushing. - Upon satisfactory completion of the sealing requirements of 3.9, the rangefinder and the end housings shall be flushed with dry nitrogen. The flushing shall continue until the gaseous mixture bled from the rangefinder and the end housings has a dew-point of at least minus 25 degrees. The instruments shall then be sealed with an internal pressure of dry nitrogen of between 0.2 and 0.5 inches of mercury.

NOTE: Correction shall be made for any pressure variation resulting from temperature and barometric changes.

3.10 Environmental requirements. -

3.10.1 Storage temperatures. - The rangefinder with the identified end housings, unattached, shall be exposed to ambient temperatures of minus 80 degrees and plus 160 degrees until thermal equilibrium is reached. After thermal equilibrium has been re-established at room temperature, the rangefinder and end housings shall show no evidence of physical failure such as cement separation, bonding failure, glass breakage, casting failure, etc., and shall meet the requirements of 3.10. Readjustment of the rangefinder and end housings will be permitted following storage temperatures, but will require that the rangefinder be completely rechecked for conformance with all post vibration requirements of 3.7.1 to 3.7.17 inclusive.

3.10.2 Operational temperatures. - The rangefinder less end housings shall be exposed to an ambient temperature of minus 40 degrees until thermal equilibrium is reached, and inspected at that temperature to insure that all knobs, locks and operational adjustments are mechanically operable. The rangefinder shall then be exposed to an ambient temperature of minus 25 degrees until thermal equilibrium is reached and then shall be inspected at this temperature for conformance with the requirements of 3.8 and table III. The rangefinder less end housings shall be exposed to an ambient temperature of plus 150 degrees until thermal equilibrium is reached and inspected at this temperature for conformance to the requirements of 3.8.

NOTE: Pre-Ordnance environmental exposure by the contractor for the purpose of stabilizing a complete instrument or any part thereof, prior to Ordnance Inspection, will be permitted and encouraged provided such treatment is part of the established process for all instruments. This process shall not be limited to those instruments selected for environmental testing.

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3.11 Post-environmental check

3.11.1 Optical check. - Any rangefinder which has been subjected to any part of the environmental requirements of 3.10 shall comply with the following requirements after thermal equilibrium has been re-established at room temperature (60°F to 90°F) with identified end housings installed.

(1) The line of sight of the left telescope through the boresight mark shall not have changed by more than 0.5 mil in deflection (azimuth) and 0.5 mil in elevation from the values in 3.7.1.1. The line of sight of the right telescope shall coincide with the aiming point of the left telescope within a maximum angular separation of 0.3 mil. Lack of horizontal coincidence of target image shall not exceed 40 UOE. Coincidence of the target images shall be re-established by using the horizontal calibration knob only. Set the horizontal calibration knob to the new median reading as obtained in preceding sentence; then, using the ICS knob, obtain a median from five ICS settings of the calibrating reticle. This median shall not differ by more than 5 UOE from median reading established in 3.7.1.4 prior to environmental.

(2) The rangefinder shall meet all requirements of 3.7.12: with the exception of the reference to paragraph 3.7.14.

(3) The field error shall not have increased from the values established in 3.7.13 by more than 2 UOE.

3.11.2 Check of production end housings. - Each end housing subjected to the applicable requirements of 3.10, shall, after thermal equilibrium has been re-established at room temperature, conform to the requirements of Ordnance Drawing F8585506.

3.12 Sealing re-check. - Rangefinder and end housings which have met the environmental requirements of 3.10 shall be rechecked to the requirements of 3.9 for leakage and re-flushed to the requirement of 3.9.1. Failure of any rangefinder or end housing to meet the requirements of 3.9 after completion of the environmental requirements of 3.10 shall not be cause for rejection of the unit provided the rangefinder or end housing is re-sealed to meet the requirements of 3.9 and 3.9.1.

3.13 Electrical check. - Electrical checks may be conducted at any time prior to final acceptance of the rangefinder.

3.13.1 Insulation resistance.- Insulation resistance between ungrounded circuits and the body or frame and between isolated circuits shall not be less than 2 meghoms.

3.13.2 Continuity. - The electrical circuits shall be checked for continuity and shall conform to the wiring diagram.

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3.14 Shock requirement. - Bonded optical assemblies, unless otherwise stated on drawing or notes, shall demonstrate ability to withstand six shocks of 150G each when shocked in accordance with the shock requirements of MIL-O-13830 by clamping or otherwise fastening the item in contact with a shock plate so that the shock is applied in a direction parallel to the bond. In the case of penta and porro reflector assemblies, the shock applied shall be parallel to the bonded and cemented surfaces and perpendicular to the bisectors at the 45 degree and 90 degree angles respectively, as formed by the aluminized surfaces.

3.15 Identification marking. - Identification marking unless otherwise specified shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.

4.1.1 Contractor inspection. - The contractor is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Contractor's quality assurance system. - The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering all components and assemblies of this specification. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when as a minimum it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.

(b) Contractor's quality standards of acceptability.

(c) Procedure for continually evaluating product quality.

(d) Procedure for continually reporting evidence of actual product quality.

The contractor will not be restricted to the inspection station or to the method of inspection listed in the written procedure provided an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure of applicable

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drawings and specifications shall be used. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that might affect the degree of assurance required by this specification and the other documents referenced herein.

4.1.3 Government verification. - All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of:

(a) Surveillance of the operations to determine whether the practices, methods, and procedures of the written inspection plan are being properly applied.

(b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until corrections have been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances.

4.1.4 Inspection equipment. - Contractor inspection referred to in par. 4.1.1 shall be performed with such inspection equipment consisting of special testing equipment, measuring instruments and gages as necessary for conformance to this specification. This equipment and maintenance thereof shall be a part of contractor's quality assurance system of 4.1.2.

4.1.4.1 The Government inspector will determine that the contractor has available, and utilizes correctly such equipment referred to in the above paragraph.

4.1.4.2 The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation on a regularly scheduled basis, and shall have available, a set of master-gages, standards, and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. Calibration equipment will be periodically checked and certified by authorized Government personnel.

4.1.4.3 Accuracy requirements. - Inspection equipment used in performance of this specification shall be capable of measuring within an accuracy of 10 percent of the specified product requirement tolerance. In cases where this requirement is seemingly inpractical, the Government shall be expediously advised in order that necessary appropriate action can be taken.

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4.1.4.4 Government equipment shall be utilized as specified under Inspection Provisions.

4.2 Initial inspection and qualifications inspection.

4.2.1 Initial inspection.

4.2.1.1 Production model specified shall be inspected by the contractor or his agent and witnessed by the Government inspector to determine acceptability in accordance with the quality assurance provisions outlined herein. Failure of the production model to comply with any of the requirements specified shall be cause for refusal by the Government to conduct a retest until corrective measures satisfactory to the Government have been taken.

4.3 Inspection provisions.

4.3.1 Lot formation. - Formation and the size of the lots submitted for acceptance inspection shall be specified by the contracting officer in accordance with Standard MIL-STD-105.

4.3.2 Examination inspection.

4.3.2.1 Components and sub-assemblies. - Classification of defects and acceptable quality levels contained in each individual SQAP listed on LSQAP 8620800 shall represent the contractor's minimum inspection requirements for those items and shall be the Government basis for acceptance or rejection by lot. The Government reserves the right to inspect for all the requirements of all applicable drawings and specifications and to reject individual non-conforming items.

4.3.2.2 Special procedures.

4.3.2.2.1 Each Rangefinder M17C submitted for acceptance inspection shall show evidence of having met the requirements of 3.3. This evidence shall be in the form of a written inspection record showing results of the positioning of Adapter Link Assemtly 8201530 in compliance with Drawing F8620801-4. This rangefinder shall consist of a main housing assembly and identified production end housings as specified.

4.3.2.2.2 Shock testing of bonded optical assemblies shall be an in-process check. Evidence of compliance with the shock requirements of 3.14 shall be presented to the Government inspector before the final acceptance of a lot of rangefinders.

4.3.2.3 Complete assembled item. - The classification of defects and acceptable quality level (AQL) listed on table IV shall constitute the contractor's minimum examination inspection requirements for the end item and shall be the Government's basis for acceptance or rejection by lot. The Government reserves the right to inspect for the requirements

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of all applicable drawings and specifications and to reject individual non-conforming items.

Table IV Classification of Defects - Examination Inspection

(Use inspection level II and sampling plan table IVA of standard MIL-STD-105)

Characteristics to be Inspected

Major:	AQL 0.65 Percent Defective	Req't Par.	Test Method or Procedure
101.	Physical failure, end housing assembly post-vibration	3.2 and 3.15	Visual
102.	Cleanliness, end housing assembly, post-vibration	3.7.2.1	Visual
103.	Base length, end housing assembly	3.7.1.6	STE
104.	Image tilt, end housing assembly	3.7.1.6	STE
105.	Elevation error, end housing assembly	3.7.1.6	STE
106.	Horizontal deviation, end housing assembly	3.7.1.6	STE
107.	Horizontal power, end housing assembly	3.7.1.6	STE
108.	Astigmatism, end housing assembly	3.7.1.6	STE
109.	Resolving power, end housing assembly	3.7.1.6	STE
110.	Horizontal calibration knob travel, pre-vibration	3.5.2.1	STE
111.	Vertical calibration knob travel, pre-vibration	3.5.2.2	STE
112.	Azimuth and elevation bore-sight knobs, pre-vibration	3.5.2.3	STE
113.	Calibration reticle setting accuracy, pre-vibration	3.5.3	STE
114.	Physical failure, main housing assembly, post-vibration	3.15 and 3.2	Visual
115.	Cleanliness, main housing assembly, post-vibration	3.7.2.1	Visual
116.	Alinement of sighting reticles	3.7.1.1	STE
117.	Coincidence of target images	3.7.1.2	STE
118.	Calibration reticle setting	3.7.1.3	STE
119.	Bright spots	3.7.2.2	Visual
120.	Parallax of sighting reticles	3.7.3.1	Visual
121.	Parallax of calibration reticles	3.7.3.2	STE

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Table IV. Cont'd

Major:	AQL 0.65 Percent Defective	Req't Par.	Test Method or Procedure
122.	Parallax of target images	3.7.3.3	STE
123.	Exit pupil	3.7.4	ME
124.	Eyepiece focus	3.7.5	STE
125.	Definition	3.7.6	ME
126.	Light transmission	3.7.7	ME
127.	Accuracy of sighting reticles	3.7.8.1	STE
128.	Plumb travel	3.7.8.2 and 3.7.8.2.1	STE
129.	Parallelism of sighting reticle and image	3.7.8.3	STE
130.	Tilt variation between reticles	3.7.8.4	STE
131.	Parallelism of calibrating reticle	3.7.9.1	STE
132.	Reversing reticles	3.7.9.2	STE
133.	Vertical movement accuracy	3.7.9.3	STE
134.	Horizontal calibration	3.7.9.4	STE
135.	ICS calibration	3.7.9.5	STE
136.	Filter error	3.7.10	STE
137.	Compensator elevation error	3.7.11	STE
138.	Range accuracy	3.7.12	STE
139.	Field error	3.7.13	STE
140.	Base length	3.7.14	STE
141.	Knob travel	3.7.15	Visual
142.	Elevation boresight knob movement	3.7.16.1	STE
143.	Deflection (azimuth) boresight knob movement	3.7.16.2	STE
144.	Backlash, boresight knobs	3.7.16.3	STE
145.	Boresight lock error	3.7.16.4	STE
146.	Boresight knob lock retention	3.7.16.5	STE
147.	Backlash, ICS and horizontal calibration knob	3.7.17.1	STE
148.	Backlash, halving and vertical calibration knob	3.7.17.1	STE
149.	Calibration lock error	3.7.17.2	Visual
150.	Operability, controls	3.8	STE
151.	Sealing, main housing and end housing	3.9	STE
152.	Insulation resistence	3.13.1	ME
153.	Continuity	3.13.2	ME

4.3.3 Inspection testing.

4.3.3.1 Control of materials. - Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or sub-contractor's facilities,

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or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.3.3.2 Environmental test of rangefinder with identified production end housing assemblies.

4.3.3.2.1 Samples for environmental testing shall be selected from acceptable rangefinders, with end housings, of the sample lots that have met all the requirements of 3.5 to 3.9 inclusive. The characteristics to be tested, and acceptable quality levels, listed in the following tables shall constitute the contractor's minimum inspection requirements for acceptance tests and will be the basis for Government acceptance and rejection by lot. The Government reserves the right to inspect for the requirements of all drawings and specifications and to reject individual non-conforming items.

Table V Inspection Testing

(Use inspection level L7 and sampling plan table IVA of MIL-STD-105)

Characteristics to be Tested

AQL 6.5 Percent Defective	Req't Par.	Test Method or Procedure
1. Controls, operability (minus 40°F)	3.10.2	Visual
2. Controls, torque (minus 25°F, plus 150°F)	3.8	STE
3. Physical failure, main housing, post-environ	3.10.1	Visual
4. Physical failure, end housing, post-environ	3.10.1	Visual
5. Base length, end housing post-environmental	3.11.2	STE
6. Image tilt, end housing, post-environmental	3.11.2	STE
7. Elevation error, end housing, post-environ	3.11.2	STE
8. Horizontal deviation, end housing, post-environ	3.11.2	STE
9. Horizontal power, end housing, post-environ	3.11.2	STE
10. Astigmatism, end housing post-environmental	3.11.2	STE
11. Resolving power, end housing, post-environ	3.11.2	STE

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Table V. Cont'd

AQL 6.5 Percent Defective	Req't Par.	Test Method and Procedure
12. Alinement of sighting reticles, post-environ	3.11.1 (1)	STE
13. Range accuracy, post-environmental	3.11.1 (2)	STE
14. Field error, post-environ	3.11.1 (3)	STE
15. Sealing, rangefinder and end housing assemblies	3.12	STE

4.3.4 Inspection equipment.

4.3.4.1 Government equipment. - Inspection equipment referenced on Index to Inspection Equipment List 8620800 will be utilized by the Government as standards for determining the adequacy of the contractor's inspection equipment in the inspection of product requirement characteristics listed on the individual inspection equipment lists. Ordnance drawings of such equipment will be made available to the bidder for informational purposes upon request.

4.3.4.2 Special test equipment to test complete rangefinder (complete rangefinder is defined as depicted on Drawing F8620800)

4.3.4.2.1 This device shall provide for testing the following requirements: Orientation, calibration reticle setting accuracy, post vibration optical re-check, alinement of sighting reticles, image, coincidence reticle setting, parallax of sighting reticles, parallax of the calibrating reticles, parallax of the target images, accuracy of sighting reticles, plumb travel, parallelism of sighting reticle and image, parallelism of calibrating reticles, filters, compensator elevation error, range accuracy, field errors, base length, boresight knobs elevation, boresight elevation, boresight knobs deflection (azimuth), boresight knobs backlash, boresight knobs locks, and post environmental optical check.

4.3.4.2.2 The device shall be equipped with suitable means to permit mounting the rangefinder, by means of the mounting bearings, in its normal operating position as depicted on Drawing F8620800. The device shall be so constructed as to permit cross-leveling within 10 seconds and the horizontal mounting surface of the device shall be horizontal and parallel with respect to a datum line (theoretically) established by the mechanical center line of the rangefinder within 25 seconds.

4.3.4.2.3 An appropriate target or targets, with acceptance and rejection limitation markings or pips marked thereon, utilized for measuring the reticle patterns shall be plumb and in line vertically and horizontally with the datum line within 25 seconds. The target or targets shall be so

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constructed as to be able to simulate target distances of from 500 meters to 4400 meters and the left line of sight shall be 90 degrees with respect to the datum line within 30 seconds. The target lines of sight shall be parallel to each other in deflection (azimuth) within 1.5 seconds and parallel to each other in elevation within 10 seconds.

4.3.4.2.4 Provisions shall be made to permit rotation of the output shaft from 500 meters to 4400 meters as indicated on the turn counter, within an accuracy of plus or minus .05 turns.

4.3.4.2.5 The device shall be equipped with a power supply capable of supplying the required power to the rangefinder during final acceptance testing.

4.3.4.2.6 An appropriate aperture plug, with an aperture diameter of .100 plus or minus .005, with a minimum movement of the aperture .030 to the left and .030 to the right, shall be provided. This aperture plug shall be utilized for testing parallax of the calibrating reticles and parallax of the target images.

4.3.4.2.7 Rigidity, stability, ease of maintenance, adaptability of set-up and calibration equipment with complete provisions are also prime requirements.

4.3.4.2.8 All special set-up and calibration equipment necessary for certification of device accuracy shall be furnished and will become a part of the device.

4.3.4.3 Special test equipment to test vibration (complete rangefinder with end housings removed)

4.3.4.3.1 The device shall be a type capable of producing vertical motion at a constant frequency of thirty cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion). This device shall be equipped with an adapter which will permit the mounting of the rangefinder in its normal operating position. The base of the adapter shall be secured firmly to the device.

4.3.4.4 Special test equipment to test torque requirements (complete rangefinder)

4.3.4.4.1 Torque wrench type testers with the following minimum ranges shall be utilized.

4.3.4.4.2 Zero to 16 pound inches, the accuracy of which shall be plus or minus 0.25 pound inches throughout the entire zone of excursion.

4.3.4.4.3 Zero to 70 ounce inches, the accuracy of which shall be plus or minus 0.25 ounce inches throughout the entire zone of excursion.

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4.3.4.4.4 These devices shall be equipped with suitable adapters to accommodate testing range knob, deflection (azimuth) boresight knobs, elevation boresight knobs, halving knob, ICS knob, horizontal calibration knob, diopter adjustment and rheostat knob, and vertical calibration knob.

4.3.4.5 Special test equipment to test end housing assembly, end housing assembly is defined as depicted on Drawing F8585506

4.3.4.5.1 This device shall provide for testing the following requirements: base length, image tilt, elevation error, horizontal deviation, horizontal power, astigmatism and resolving power.

4.3.4.5.2 The device or devices shall permit mounting the end housing assembly in its normal operating position, as depicted on Drawing F8585506, so that surface "X" is in complete contact with the vertical mounting surface of the device. The qualification surfaces shall be in complete contact with the corresponding mating surface of the device. The qualification mating surface shall be plumb within 1 minute.

4.3.4.5.3 An appropriate target with acceptance and rejection limitation markings or pips marked thereon shall be provided.

4.3.4.5.4 Rigidity, stability, ease of maintenance and adaptability of set-up and calibration equipment with complete provisions are also prime requirements.

4.3.4.5.5 All special set-up and calibration equipment necessary for certification of device or devices accuracy shall be furnished and will become a part of the device.

4.3.4.6 Special test equipment to test vibration, end housing assembly.

4.3.4.6.1 The device shall be a type capable of producing vertical motion at a constant frequency of thirty cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion). This device shall be equipped with an adapter which will permit the mounting of the end housing assembly in its normal operating position. The base of the adapter shall be secured firmly to the device.

4.4 Test methods and procedures.

4.4.1 Identified production end housing assembly (less main housing).-

4.4.1.1 Vibration. - The identified end housing assembly shall be mounted in its normal operating position upon special testing equipment and vibrated in accordance with the requirements of 3.6.1.

4.4.1.2 Physical failure post-vibration.- As a result of vibration, the end housing assembly shall show no signs of physical failure of mechanical or optical parts.

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4.4.1.3 Cleanliness, post-vibration. - The end housing assembly shall be inspected for cleanliness in accordance with Specification MIL-O-13830 and shall meet the requirements of 3.7.2.1.

4.4.1.4 Post-vibration test of end housing assembly. - Subsequent to vibration and prior to assembly to the rangefinder main housing, the end housing assembly subjected to that test shall meet the requirements of Drawing F8585506. The end housing assembly shall be positioned on a device or devices, special testing, appropriate for measuring base length, image tilt, elevation error, horizontal deviation, horizontal power, astigmatism, and resolving power.

NOTE: It is recommended that the special test equipment be set-up and oriented by use of a calibrated end housing assembly.

4.4.1.5 The failure of the identified end housing to meet the above requirements shall be cause for rejection of that rangefinder with which they are identified.

4.4.1.6 Mounting of the end housing assemblies on the rangefinder main housing shall be in compliance with the requirements on Drawing F8620800.

4.4.2 Orientation of rangefinder with end housing for tests, pre-vibration. The requirements set forth in 3.5.1 shall be met with the Rangefinder M17C mounted on special test equipment with a line through the centers of the 2 mounting bearings horizontal. The geometric centers of the 2 mounting bearings and the bore on Adapter Assembly 8201530 shall be in a plane perpendicular to the horizontal line within the tolerance specified on Drawing F8620800. A line of sight through the rangefinder shall be directed to a calibrated target, positioned at the specified distance, appropriate for measuring the accuracy requirements of the rangefinder. The horizontal target line shall be coincident to the line through the centers of the 2 mounting bearings. The vertical target line shall be perpendicular to that line through the centers of the mounting bearings.

4.4.3 Knob travel - pre-vibration.

4.4.3.1 Horizontal calibration knob. - When the horizontal calibration knob has been set to the median coincidence reading (coincidence of left and right target image) established in 3.5.1 (9), the knob shall be at the mid-point of rotation within the tolerance specified in 3.5.2.1. Rotation in UOE displaced from initial established mid-point shall be read from knob scale. Slip scale to aline zero index to fixed upon successful completion of test.

4.4.3.2 Vertical calibration knob. - When vertical coincidence of the left and right target image has been established as specified in 3.5.1 (8), the vertical calibration knob shall be at the mid-point of rotation within the tolerance specified in 3.5.2.2. Rotation from initial established mid-point shall be the measured distance between the fixed and moveable index lines. Slip scales to aline index lines upon successful completion of test. Mark position of knob for subsequent test.

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4.4.3.3 Deflection (azimuth) and elevation boresight knobs. - The deflection (azimuth) and elevation boresight knobs shall be set to their mid-point of travel as specified in 3.5.1 (10), and the scales slipped to aline the deflection (azimuth) three (3) and elevation two (2) graduation line to the fixed index. When the line of sight through the boresight mark (center of boresight cross) of the left and right sighting reticle is aligned to the positions specified in 3.5.1 (11) and 3.5.1 (12), the deflection (azimuth) and elevation boresight knobs shall be at their mid-point of rotation within the tolerance specified in 3.5.2.3 as measured on the boresight scales. Slip scale to aline the deflection (azimuth) three (3) and elevation two (2) graduation line to the fixed indexes upon successful completion of test.

4.4.4 Calibration reticle setting accuracy, pre-vibration. - With the rangefinder oriented as in 3.5.1 and the calibration knobs set to mid-point within the tolerance specified, the lower calibration reticle shall meet the requirements of 3.5.3.1 as measured on a target appropriate for measuring displacement of the calibration reticle with respect to the sighting reticle.

4.4.4.1 The upper calibration reticle shall be adjusted in elevation by means of the halving knob, to meet the requirements of 3.5.3.1. Mark position of halving knob for subsequent test.

4.4.4.2 The ICS knob shall move the upper calibration reticle in azimuth to vertically aline the calibration reticles. A group of five horizontal coincidence settings of the calibration reticles shall be accomplished by means of the ICS knob. The median of this group of settings shall not exceed the tolerance specified in 3.5.3.2 as read on the ICS knob scale.

4.4.5 Vibration.

4.4.5.1 Vibration, main housing assembly. - The rangefinder shall be mounted in its normal operation position upon special test equipment and vibrated in accordance with the requirements of 3.6.

4.4.5.2 Physical failure, main housing assembly. - As a result of vibration, the main housing assembly shall show no signs of physical failure of mechanical or optical parts.

4.4.5.3 Cleanliness, main housing assembly, post-vibration. - The rangefinder, less end housings, shall be inspected for cleanliness in accordance with Specification MIL-O-13830, and shall meet the requirements of 3.7.2.1.

4.4.6 Subsequent to vibration, the rangefinder, with identified end housing assemblies installed, shall be set up as specified in 4.4.2. Certain tests shall require auxiliary testing equipment to be used in conjunction with the special test equipment specified in 4.4.2 for determination of compliance with the requirements.

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4.4.6.1 Alinement of sighting reticles, post-vibration. - The line of sight through the boresight mark of the left and right sighting reticles shall again be alined to the positions specified in 3.5.1 (11) and 3.5.1 (12). Any change in the alinement of the sighting reticles from the values established in 4.4.3.3 shall not exceed the tolerance specified in 3.7.1.1 as read on the knob scale.

4.4.6.2 Coincidence of target image, post-vibration. -

4.4.6.2.1 When the vertical calibration knob is rotated to the position established in 4.4.3.2, vertical coincidence of the target images shall not have changed by more than the tolerance specified in 3.7.1.2 as read on a target appropriate for measuring displacement of target images.

4.4.6.2.2 The median of five readings to bring the target images into horizontal coincidence, by means of the horizontal calibration knob, shall not differ from the zero setting established in 4.4.3.1 by more than the tolerance specified in 3.7.1.2. Any change in UOE shall be read on the horizontal calibration knob scale. Slip scale to the new zero upon successful completion of test.

4.4.6.3 Calibration reticle setting.

4.4.6.3.1 Subsequent to vibration, the space between the two calibration reticles shall meet the requirements of 3.7.1.3 when the vertical calibration knob is rotated to the position established in 4.4.3.2 and the halving knob is rotated to the position established in 4.4.4.1.

4.4.6.3.2 The horizontal calibration knob shall be set to the median reading obtained in 4.4.6.2.2. A group of five horizontal coincidence settings of the calibration reticles shall be accomplished by means of the ICS knob. The difference between the median of this group and the value established in 4.4.4.2 shall not exceed the tolerance specified in 3.7.1.4.

4.4.6.4 Bright spots. - The reticles, illuminated by the lamps in the rangefinder, shall be visually inspected for conformance with 3.7.2.2.

4.4.6.5 Parallax of sighting reticles. - The test shall be conducted with each sighting reticle projected into the optical system, in turn. Parallax at the reticle boresight mark for each optical system shall not exceed the tolerance specified in 3.7.3.1. When the image of the left and right sighting reticles are superimposed, the parallax between the reticles shall not exceed the tolerance specified in 3.7.3.1. Parallax is any apparent movement of the reticles in relation to each other or to the target in the field of view caused by movement of the observer's head.

4.4.6.6 Parallax of the calibration reticles. - The test shall be conducted with the aid of an auxiliary testing device containing a central aperture that will permit controlled scanning of the exit pupil. This device shall be positioned over the rangefinder eyepiece. A group of five

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horizontal coincidence settings of the calibration reticles shall be accomplished by means of the ICS knob when the eye is positioned the specified distance to the left of exit pupil center. A group of five readings shall be taken with the eye positioned to the right of exit pupil center. The difference between the median of the left group and the median of the right group shall not exceed the tolerance specified in 3.7.3.2 as read on the ICS knob scale.

4.4.6.7 Parallax of the target images. - The auxiliary testing device described in the test for parallax of calibration reticles shall be used for this test. The images of a target at 1500 meters plus or minus 100 meters shall be brought into coincidence by means of the range knob. Coincidence shall then be disturbed by rotation of the horizontal calibration knob. A group of five horizontal coincidence settings of the target image shall be accomplished by means of the horizontal calibration knob when the eye is positioned the specified distance to the left of exit pupil center. A group of five readings shall be taken with the eye positioned to the right of exit pupil center. The difference between the median of the left group and the median of the right group shall not exceed the tolerance specified in 3.7.3.3 as read on the horizontal calibration knob scale.

4.4.6.8 Exit pupil. - This test shall be conducted in accordance with Specification MIL-O-13830 and shall meet the requirements of 3.7.4. Exit pupil measurement shall be made with end housing assemblies installed on the rangefinder.

4.4.6.9 Eyepiece focus.

4.4.6.9.1 The diopter movement of the eyepiece shall be tested manually and visually for the requirements of 3.7.5.1.

4.4.6.9.2 Focus of the rangefinder eyepiece shall be tested with the aid of a calibrated dioptometer, having a magnification of at least three-power (3x), adjusted for infinity focus for the observer's eye, and the diopter scale set to zero. The zero setting of the rangefinder eyepiece diopter scale shall be within the tolerance specified in 3.7.5.2 when sharp focus of the left target image is obtained.

4.4.6.10 Definition. - This test shall be conducted in accordance with Specification MIL-O-13830. The resolving power of each system shall meet the requirements of 3.7.6. The auxiliary telescope used in this test shall be at least a three-power (3x) instrument.

4.4.6.11 Light transmission. - Filters shall be transferred out of the optical systems for this test. Light rays from a concentrated source entering the rangefinder, through each end housing window, shall be measured at the eyepiece by means of a photometer, consisting of a light source, photocell and micrometer. The difference in light transmission between the two optical systems shall meet the requirements of 3.7.7.

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4.4.6.12 Sighting reticles.

4.4.6.12.1 Accuracy of sighting reticles. - This test shall be conducted in accordance with Specification MIL-O-13830. The angle subtended by the total horizontal lines and spaces of the left and right sighting reticles shall meet the requirements of 3.7.8.1.

4.4.6.12.2 Plumb travel. - A line of sight through the boresight mark of the left sighting reticle coincident to a plumb line shall not deviate from that plumb line by more than the tolerance specified in 3.7.8.2 when the rangefinder is manually elevated from 8 degrees depression to 19 degrees elevation.

4.4.6.12.2.1 Rangefinder movement. - Rangefinder shall be elevated and depressed to meet the requirements of 3.6.8.2.1.

4.4.6.12.3 Parallelism of sighting reticle and image. - Parallelism of the longest horizontal reticle line to the image of a horizontal target line shall be measured on a target appropriate for measuring any cant of the reticle line. Parallelism of the left and right sighting reticle shall meet the requirements of 3.7.8.3.

4.4.6.12.4 Tilt variation between reticles. - When the right sighting reticle is superimposed on the left sighting reticle, the tilt variation between reticles, as measured at one extremity of the 40 mil reticle line, shall not exceed the tolerance specified in 3.7.8.4, when the other extremity of both 40 mil reticle lines are coincident. Tilt between reticles shall be measured on a target appropriate for measuring tilt variation between reticles within the tolerance specified.

4.4.6.13 Coincident reticles.

4.4.6.13.1 Parallelism of Calibrating reticles. - Each calibrating reticle, in turn, shall be tested for parallelism to the image of a plumb line and shall be parallel within the tolerance specified in 3.7.9.1. When one extremity of a vertical side of the calibration reticle is alined coincident to the image of a plumb line, the other extremity shall be parallel to that plumb line within the width of the calibration reticle at that extremity.

4.4.6.13.2 Reversing reticles. - With the calibration reticles in reversed position (upper bar immediately under the lower bar), a group of five horizontal coincidence settings of the calibration reticles shall be accomplished by means of the ICS knob. The difference between the median of this group and the value established in 4.4.6.3.2 shall not exceed the tolerance specified in 3.7.9.2 as read on the ICS knob scale.

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4.4.6.13.3 Upon successful completion of the vertical movement accuracy test, restore the calibration reticles to their proper position.

4.4.6.13.4 Vertical movement accuracy. - Set the horizontal calibration knob and the ICS knob to the median values obtained in 4.4.6.3.2. The vertical calibration knob shall be rotated to the maximum CCW position and the upper calibration reticle, controlled by the halving knob, shall be positioned immediately above the lower reticle (separated by a minimum distinguishable spacing). A group of five horizontal coincidence settings of the calibration reticle shall be accomplished by means of the ICS knob. The vertical calibration knob shall then be rotated to the maximum CW position and the upper calibration reticle again positioned as above. A group of five horizontal coincidence settings of these reticles shall be accomplished by means of the ICS knob. The difference between the median of the first group and the median of the second group shall not exceed the tolerance specified in 3.7.9.3 as read on the ICS knob scale. Upon successful completion of this test, return calibration reticles to normal position.

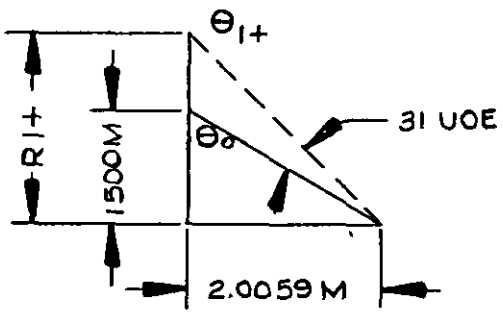
4.4.6.13.5 Horizontal calibration. - The line of sight of the rangefinder shall be directed to a target of known distance within 1400 and 1600 meters and range scale set to agree with the range chosen. The target shall be in coincidence and the horizontal calibration and ICS knobs set to established median positions. The horizontal calibration knob shall be rotated plus 40 UOE, then coincidence of the target images re-established by means of the ranging knob. The corresponding change in range, as read on the range scale, shall be equivalent to the change in the number of UOE within the tolerance specified in 3.7.9.4. This test shall be repeated with the horizontal calibration knob rotated to minus 40 UOE. In addition to the 10 percent tolerance (4 UOE) allowed in 3.7.9.4, five UOE are allowed to cover the error in the positioning of the horizontal calibrating wedge relative to the ranging plane and its angular tolerance. Therefore, the change in range could correspond to 40 UOE plus or minus 9 UOE.

An example, using a target at 1500 meters, follows:

The calculations must be made for 31 UOE and 49 UOE in both directions from the 1500 meter position.

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Increasing Range



$\tan \theta_0 = \frac{2.0059}{1500} = .0013372$

$\theta_0 = .0766°$

31 UOE = 31 x 1.2 Sec.
31 UOE = 37.2 Sec.

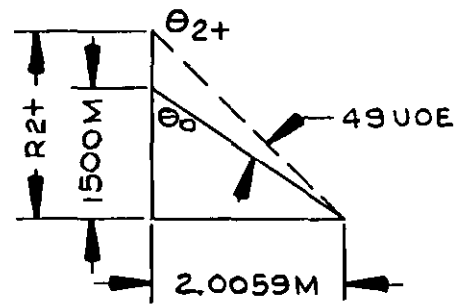
$37.2 \text{ Sec.} = \frac{37.2}{3600} = .0103°$

$\theta_{1+} = \theta_0 - .0103°$
$\theta_{1+} = .0766 - .0103$

$\theta_{1+} = .0663°$

$\tan \theta_{1+} = .0011571$

$R_{1+} = \frac{2.0059}{\tan \theta_{1+}}$
$R_{1+} = 1733.5 \text{ meters}$



$\tan \theta_0 = \frac{2.0059}{1500} = .0013372$

$\theta_0 = .0766°$

49 UOE = 49 x 1.2 Sec.
49 UOE = 58.8 Sec.

$58.8 \text{ Sec.} = \frac{58.8}{3600} = .0163°$

$\theta_{2+} = \theta_0 - .0163°$
$\theta_{2+} = .0766 - .0163$

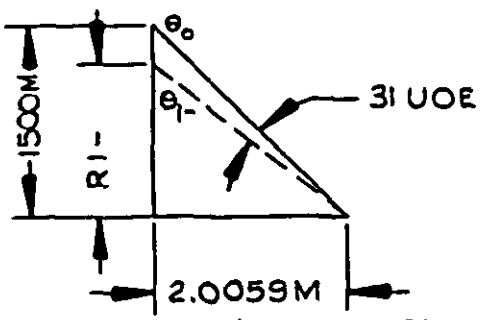
$\theta_{2+} = .0603°$

$\tan \theta_{2+} = .0010524$

$R_{2+} = \frac{.20059}{\tan \theta_{2+}}$
$R_{2+} = 1906.0 \text{ meters}$

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Decreasing Range



$\tan \theta_0 = \frac{2.0059}{1500} = .0013372$

$\theta_0 = .0766°$

31 UOE = 31 x 1.2 Sec.
31 UOE = 37.2 Sec.

37.2 Sec. $= \frac{37.2}{3600} = .0103°$

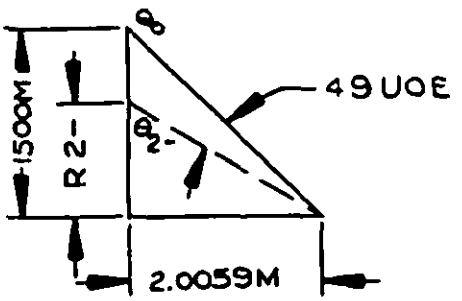
$\theta_{1-} = \theta_0 + .0103° = .0766 + .0103$

$\theta_{1-} = .0869°$

$\tan \theta_{1-} = .0015167$

$R_{1-} = \frac{2.0059}{\tan \theta_{1-}}$

$R_{1-} = 1322.5$ meters



$\tan \theta_0 = \frac{2.0059}{1500} = .0013372$

$\theta_0 = .0766°$

49 UOE = 49 x 1.2 Sec.
49 UOE = 58.8 Sec.

58.8 Sec. $= \frac{58.8}{3600} = .0163°$

$\theta_{2-} = \theta_0 + .0163°$

$\theta_{2-} = .0929°$

$\tan \theta_{2-} = .0016214$

$R_{2-} = \frac{2.0059}{\tan \theta_{2-}}$

$R_{2-} = 1237.1$ meters

Range scale reading must lie within the following limits:

1237 meters	1733 meters
1322 meters	1906 meters

4.4.6.13.6 ICS calibration. - The horizontal calibration knob and the ICS knob shall be set to the values established in 4.4.6.3.2. The ICS and horizontal calibration knobs shall be rotated to the positions specified in 3.7.9.5 as measured on the knob scales. The position of the horizontal calibration knob, as a result of this excursion, shall not exceed the tolerance specified in 3.7.9.5 as read on the horizontal calibration knob scale.

4.4.6.14 Filters.

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4.4.6.14.1 Introduction of the filters shall not cause the target images to shift horizontally with respect to each other by more than the tolerance specified in 3.7.10. Image shift shall be measured on a target appropriate for measuring image shift within the specified tolerance.

4.4.6.14.2 Introduction of the filters shall not cause the target images to shift vertically with respect to each other by more than the tolerance specified in 3.7.10. Image shift shall be measured on a target appropriate for measuring image shift within the specified tolerance.

4.4.6.15 Compensator elevation error. - The rangefinder range scale shall be set to 500 meters with the target images in vertical coincidence. Any vertical shift of the target images shall not exceed the tolerance specified in 3.7.11 when the range scale is actuated from 500 meters to 4400 meters. Image shift shall be measured on a target appropriate for measuring image shift within the specified tolerance.

4.4.6.16 Range accuracy. - The horizontal calibration knob shall be set to the value established in 4.4.6.3.2. An auxiliary testing device that shall be capable of accurately indicating rotational positions of the rangefinder output shaft within the tolerances specified in table I of 3.7.12.2, when the range scale is set to any range between 500 meters and 4400 meters shall be attached to the rangefinder output shaft. This testing device shall indicate an output shaft rotational value (P) of 15.92 when the range scale indicates 1500 meters of range.

4.4.6.16.1 The range of rotation of the output shaft shall meet the requirements of 3.7.12.1 as measured on the testing device.

4.4.6.16.2 The rangefinder line of sight shall be directed to a target or targets positioned at known distances between 500 - 540 meters, 650 - 750 meters, 900 - 1100 meters, 1400 - 1600 meters, 2200 - 2800 meters and 3800 - 4400 meters. The "P" value for the known range shall be computed from the formula specified in 3.7.12 and shall be computed for each target. The median of each group of five range readings (rotational position of output shaft) shall meet the requirements of 3.7.12.2 within the UOE tolerance specified in table I. Unit of error tolerances shall be converted to output shaft rotations using the formula specified in 3.6.12.2 and compared with the difference between the indicated output shaft value of the median range reading and the computed "P" value for a specific range.

4.4.6.16.3 When the rangefinder output shaft is rotated to the nominal positions (P value) specified for the ranges in 3.7.12.3, column I, table II, the range scale shall indicate these ranges respectively within the tolerance specified. The nominal positions of the output shaft shall be computed from the formula specified in 3.7.12.

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4.4.6.17 Field errors. - This test shall require five targets to be positioned in a vertical plane at 1500 meters plus or minus 100 meters. The rangefinder scale shall be set to agree with the range chosen. One target shall be positioned at the center of the field (a line of sight through the boresight mark) with the remaining four targets positioned at the distance from the center of the field specified in 3.7.13. The target at the center of the field shall be in coincidence when the horizontal calibration and ICS knobs are set to their established median positions. A group of five horizontal coincidence settings shall be taken on each target, in turn, by means of the horizontal calibration knob. The difference between the median of the center target and the medians of each group of five settings shall not exceed the tolerance specified in 3.7.13 as measured on the horizontal calibration knob scale.

4.4.6.18 Base length. - All control knobs of the rangefinder shall be set to their individual established median positions and the range scale set to 4400 meters of range. A plumb line shall be positioned approximately 3 to 4 inches in front of each end housing window. The measured distance between plumb lines shall be 79.000 inches plus or minus 0.15 inches (2.0059 meters plus or minus 3.81 millimeters). The test for base length shall be performed with the aid of an auxiliary testing device, positioned at the eyepiece of the rangefinder. This device shall be capable of focusing on the etched dots on Objectives 8200012, 8200005, and the plumb lines. The scale or reticle of the device shall be capable of measuring the actual separation between dot images and plumb line images formed at the rangefinder eyepiece. Focus the device on the etched dots, then turn occluder of rangefinder so that left dot image is observed. Record scale reading of position of left dot. Turn occluder so that right dot image is observed. Record scale reading of position of right dot. The difference between the two recorded readings shall be used as the "A" factor in the formula for computing base length error. Repeat the operations listed above to measure the difference between plumb lines. The difference between the two recorded readings shall be used as the "B" factor in the following formula for computing base length error where 10 equals the power of the rangefinder.

$$10(A-B) = \text{Base length error}$$

The base length error shall not exceed the tolerance specified in 3.7.14.

4.4.6.19 Knob travel. - All knobs shall be visually and manually inspected for conformance with 3.7.15.

4.4.6.20 Boresight knobs.

4.4.6.20.1 Elevation. - The lines of sight, through the boresight mark of the left and right sighting reticle, shall be oriented to the position established in 4.4.3.3. Each mil of rotation of the elevation boresight knobs, clockwise and counterclockwise from the established nominal position, shall cause a comparable change in the line of sight and

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shall meet the excursion requirements of 3.7.16.1 as measured on a target appropriate for measuring elevation movement of the reticle.

4.4.6.20.2 Deflection (azimuth). - The line of sight, through the boresight mark of the left and right sighting reticle, shall be oriented to the position established in 4.4.3.3. Each mil of rotation of the deflection (azimuth) boresight knobs, clockwise and counterclockwise from the established nominal position, shall cause a comparable change in the line of sight and shall meet the excursion requirements of 3.7.16.2 as measured on a target appropriate for measuring deflection (azimuth) movement of the reticle.

4.4.6.20.3 Backlash boresight knobs. - Rotate a line of sight, through the boresight mark of the sighting reticle, in a continuous clockwise direction by means of the deflection (azimuth) boresight knob until the vertical reticle line is in coincidence with a vertical target line. Record setting of knob scale. Rotate line of sight past the vertical target line (clockwise) a minimum of one full turn of the deflection (azimuth) boresight knob, then by a continuous counterclockwise rotation of the azimuth knob bring the vertical reticle line into coincidence with the vertical target line and read the knob scale. The difference between the recorded reading and the second reading is the amount of backlash. Repeat this operation for the elevation boresight knob using the horizontal reticle line and a horizontal target line. Backlash for each elevation on each deflection (azimuth) knob shall hot exceed the limit specified in 3.7.16.3.

NOTE: Rotation of the knobs must be uniform in one direction. The reticle line shall not override the target line during these operations. If override occurs, this test is nullified and the operations specified above will again be performed.

4.4.6.20.4 Boresight locks. - This test may be conducted concurrent with the aforementioned test. Actuation of the deflection (azimuth) and elevation boresight knob locks shall not cause the line of sight to shift nor the boresight knobs to rotate more than the tolerance specified in 3.7.16.4.

4.4.6.20.5 Boresight knob lock retention. - This test shall be performed with a device or devices, special testing, affixed to the boresight knobs that shall be capable of measuring applied torques. The boresight knob lock retention shall meet the requirements of 3.7.16.5 when the specified torque is applied to the knobs.

4.4.6.21 Calibration knobs.

4.4.6.21.1 Backlash in ICS knob and horizontal calibration knob.- These tests shall be conducted similar to 4.4.6.20.3. The upper calibration reticle shall be alined to a vertical target line by means of the ICS knob. The clockwise rotation of the knob shall be to the clockwise stop before

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beginning the counterclockwise rotation. The lower calibration reticle shall be alined to the vertical target line by means of the horizontal calibration knob. The results of this test shall meet the requirements of 3.7.17.1.

4.4.6.21.2 Backlash in halving knob and vertical calibration knob. - These tests shall be conducted similar to 4.4.6.20.3. The upper calibration reticle shall be alined to a horizontal target line by means of the halving knob. The lower calibration reticle shall be alined to the horizontal target line by means of the vertical calibration knob. The results of this test shall meet the requirements of 3.7.17.1.

4.4.6.21.3 Calibration locks. - This test may be conducted concurrent with the backlash test for calibration knobs. Any error caused by the engagement of the horizontal and vertical calibration knobs with the serration detents shall not exceed the tolerance specified in 3.6.17.2.

4.4.7 Operability. - The maximum torques required to rotate the controls of the rangefinder shall not exceed those values specified in column I, table III, of 3.8 while at an ambient temperature of plus 60 to 90 degrees F . These tests shall be conducted with special test equipment capable of measuring the applied torques.

4.4.8 Sealing and flushing of rangefinder main body and end housing assembly.

4.4.8.1 The rangefinder body and end housing assembly shall be flushed with dry nitrogen. The rangefinder body and end housing assembly shall then be sealed with an internal pressure of 6 inches of mercury. After a period of 12 hours, the body and end housing shall show no evidence of leakage of dry nitrogen in excess of the tolerance specified in 3.9. The sealing test shall be conducted with special test equipment that shall be capable of measuring any drop in internal pressure of rangefinder body and end housing assembly within the tolerance specified.

4.4.8.2 Upon satisfactory completion of the sealing test, the rangefinder body and end housing assemblies shall be bled to the internal pressure specified in 3.9.1. At this pressure, the dewpoint temperature shall meet the requirements of 3.9.1 as measured on a dew-point tester.

4.4.9 Insulation resistance. - This test shall be conducted in accordance with Specification MIL-F-13926 by use of standard measuring equipment and shall meet the requirements of 3.13.1. Remove all lamp bulbs and remove ground connection, identified as wire number 17 on Drawing 8620816 before conducting this test.

4.4.10 Continuity. - The electrical circuits shall be checked for continuity by use of standard measuring equipment and shall conform to the wiring diagram.

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4.4.11 Environmental test of rangefinder with identified production end housing assemblies.

4.4.11.1 Environmental test shall be performed using the contractor's Government approved refrigeration and heating equipment.

4.4.11.2 The operability and torque test on the rangefinder may be conducted concurrent with the storage test. Caution must be exercised so as not to expose the rangefinder and end housing assemblies to thermal shock during environmental test. Temperature changes shall be gradual.

4.4.11.3 Controls, operability. - When thermal equilibrium is reached at minus 40 degrees F, the controls listed in table III of 3.8 shall be mechanically operable.

4.4.11.4 Controls, torque. - When thermal equilibrium is reached at minus 25 degrees F and plus 150 degrees F, all controls shall meet the applicable torque requirements of 3.8.

4.4.11.5 Physical failures, main housing and end housing assemblies, post-environmental. - After thermal equilibrium has been reached at the minus and plus temperatures specified in 3.10.1, the rangefinder and end housing assemblies shall be returned to room temperature and examined for evidence of physical failure of optical and mechanical parts. Failure of rangefinder or either end housing assembly to meet this test shall be cause for rejection of the rangefinder assembly.

4.4.11.6 Post-environmental test of rangefinder with installed end housing assemblies. - Subsequent to the test specified in 4.4.11.4 and 4.4.11.5, the rangefinder shall be set up as specified in 4.4.2. The boresight knobs and calibration knobs shall be set to their established values. The rangefinder shall then meet the requirements of 3.11.1.

4.4.11.6.1 Alinement of sighting reticles. - The line of sight through the boresight mark of the left sighting reticle shall meet the requirements of 3.11.1 (1). The line of sight through the boresight mark of right sighting reticle shall coincide with the aiming point of the left telescope within the tolerance specified in 3.11.1 (1). Any adjustment necessary to meet this requirement shall be made with the horizontal calibration knob and shall not exceed the UOE tolerance specified, as read on the knob scale. The UOE tolerance specified is not a median reading.

4.4.11.6.2 Range accuracy. - This test shall be conducted as specified in 4.4.6.16, and shall meet the requirements of 3.7.12.

4.4.11.6.3 Field error. - This test shall be conducted as specified in 4.4.6.17, and shall meet the requirements of 3.11.1 (3).

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4.4.11.7 Post-environmental, sealing recheck of the rangefinder and end housing assemblies. - Subsequent to environmental test, the rangefinder and end housing assemblies subjected to that test shall again meet the requirements of 3.9. This test shall be as specified in 4.4.8. Failure of any rangefinder or end housing to meet the requirements of 3.9 shall not be cause for rejection provided the rangefinder is re-sealed and flushed to meet the requirements of 3.9. Disassembly of the rangefinder to re-seal shall not be permitted.

4.4.12 Final acceptance. - Final acceptance of the lot of rangefinders (with identified end housings) shall be withheld by the Government inspector until the sample lots of rangefinders (with identified end housings), representative of that production lot has satisfactorily passed the applicable requirements of this detailed specification.

4.4.13 The rangefinder with identified end housings shall be shipped as a complete unit.

4.4.14 Workmanship. - Exclusive of the characteristics specified in the classification of defects, the rangefinder shall be free from any damage, defect, or contamination, the nature, degree or extent of which in the judgement of the Government inspector, detrimentally affects the appearance or serviceability of the intended use of the rangefinder.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Specification FCP-4051 and instructions forming part of the contract or furnished by the contracting officer in connection with the contract.

6. NOTES

6.1 Intended use. - The Rangefinder M17C is a coincidence image type instrument, intended for installation in the 90MM Gun, Tank M48A2C and 105MM Gun, Tank M60. This rangefinder is a commander operated, it employs a 10 power optical system, with a 4 degree field of view and has a base length, including end housings of 79 inches (2.0059 meters) and furnishes range data to a trajectory computing system.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of an applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Provisions for production model including date of delivery (see 3.1).

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6.3 Design information. - Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon specific request of the contractor.

6.4 Definition.

6.4.1 Median. - The term "median" as used herein, is defined as that range or setting which divides the series into an upper and lower half. Example - in a set of five range readings 2000, 2050, 2080, 2060, 2025 the median would be 2050. Two ranges are higher and two are lower.

6.4.2 Special test equipment (STE). - Special inspection fixtures, instruments, and modified standard measuring equipment and special purpose commercially available equipment.

6.4.3 Standard measuring equipment (SME). - Standard measuring equipment includes the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement, that is, micrometer, height gages, gage blocks, depth gauges, scale protractors, dial indicator, voltmeters, ohmmeters, etc.

6.5 Pilot lot. - A pilot when required shall be specified in the invitation for bid. In the absence of a specific requirement, preparation of a pilot lot shall be at the discretion of the contracting officer.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other then in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

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